Jev-Grounded Adaptive Temporal Granularity for Causal Video Generation: An Audit of the Cost, Signal and Endpoint Preconditions

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***Abstract*—Adaptive temporal granularity — deciding when a causal video generator should produce one frame at a time (*fine*) versus several frames per chunk (*coarse*) — rests on four preconditions: a real mode-dependent cost gap, a pre-generation signal, an endpoint independent of the compared modes, and an oracle calibrated well enough to gate on. This report audits all four with an off-device, image-grounded System One oracle (Jev 1.14) on a matched factorial over two accelerators (Tesla T4 and A100-80GB). We contribute: (i) a *break-even law* that is conditional on both the hardware and the operating point, which resolves a confound in the earlier audit of this problem; (ii) a paired test showing that an *image*-grounded oracle is a *worse* routing signal than a text-grounded one; (iii) a validated independent *endpoint* — the image-grounded oracle agrees with CLIP subject consistency at Spearman $\rho{=}+1.000$; and (iv) two honest negatives (a confidence-gated policy that is neutral, and a stability-gated recalibration that is worse on both GPUs). Every headline number is re-derived from a stored artifact (18/18 claim-to-artifact checks pass). The reusable output is the break-even law plus the endpoint; the routing signal remains an open problem.**

***Index Terms*—adaptive granularity, temporal granularity, causal video generation, video world models, autoregressive diffusion, off-device oracle, System One, efficient inference.**

I. Introduction

Causal (autoregressive) video generators can trade computation for quality by changing their *temporal granularity*: frame-wise generation is accurate but expensive, while chunk-wise generation is cheaper on some hardware but drifts. A natural system spends fine computation only on hard segments, deciding before each segment is generated. That system is only viable if four things hold, and this report measures each of them.

a) Contributions.:

1) **A matched break-even law** (§V-A) over $\{\text{T4}, \text{A100}\} \times \{\text{fine } s_1/s_2/s_4\} \times \{\text{coarse } s_1/s_2/s_4\}$. Coarse is cheaper *iff* the measured per-forward ratio falls below the schedule break-even, and the same pair can sit on opposite sides of that inequality on the two GPUs. This resolves a confound in the earlier audit [1], which compared an *expensive* coarse mode on T4 with a *cheap* one on A100.
2) **A paired modality test** (§V-B). On 80 matched segments, an image-grounded oracle is a *worse* routing signal than a text-grounded one (ΔAUROC -0.110 to -0.195).
3) **A validated independent endpoint** (§V-E). The same image-grounded oracle, used as a quality endpoint, agrees with CLIP subject consistency at Spearman $\rho{=}+1.000$.
4) **Two honest negatives** (§V-G, §V-H): confidence-gated routing is neutral, and stability-gated recalibration is worse on both GPUs.

b) Relationship to the prior study.: The prior preprint [1] framed the problem and reported a conditional trade-off on one GPU. This report is the audit that follows: it measures the preconditions on two GPUs with a matched design, tests the modality of the oracle, and validates the endpoint. The companion method (DDAG) is a separate artifact; neither reuses text or figures from the prior preprint.

II. Related Work

a) Causal video generation.: Streaming generators roll out chunk by chunk, and closing the train–test gap between teacher forcing and self-rolling is central [2], [3]. Few-step causal distillation makes the setting interactive [4]; we treat such a model as fixed and study its granularity.

b) Adaptive computation.: Adaptive computation was introduced for recurrent networks [5] and early exiting for classification [6], [7]. In generative video, computation is allocated by step [8], by chunk shape [9], by pruning and routing [10], and by layer reuse [11]. Our object is the *temporal chunk size* of a fixed backbone, driven by an *off-device* oracle.

c) Evaluation.: Video generators are evaluated along many axes [12], [13]; we adopt CLIP subject consistency [14] and a captioner [15] as independent endpoints, and compare an image-grounded decision-model endpoint against them.

d) Decision models.: System One decision models return typed, calibrated answers [16]; independent evidence suggests pixel-only quality prediction with such models is unreliable [17], which is consistent with our signal result.

III. Preconditions and Method

a) Preconditions.: **P1 (cost)** coarse must be cheaper than fine *on the deployment hardware and at the chosen operating point*. **P2 (signal)** a quantity computable *before* a segment is generated must identify coarse-unsafe segments. **P3 (endpoint)** quality must be certified by a measure independent of the compared modes. **P4 (calibration)** the oracle's confidence must be informative enough to gate on.

b) Cost model.: With t_c, t_f the seconds per single forward and NFE the number of forward evaluations per segment, coarse is cheaper iff

$$\frac{t_c}{t_f} < \frac{\mathrm{NFE}_f}{\mathrm{NFE}_c} \equiv \rho_{\mathrm{be}}, \tag{1}$$

i.e. the *measured per-forward ratio* must fall below the *schedule break-even*.

c) Oracle.: Jev 1.14 is queried off-device. A *text* state passes a caption plus motion statistics; an *image* state passes the last three context frames as assets in a `multipart/form-data` request with the header `TypeSafe-Danger-Zone-Images: true`. The oracle returns typed answers (a difficulty score, a drift risk, a motion class) and a confidence.

d) Backbone, modes and design.: Backbone: Causal-rCM Wan2.1-T2V-1.3B (commit `ed3cb14`) [4]. Modes: fine `c1-1` (s steps), cheap coarse `c3-3 s1`, expensive coarse `c3-3 s4`, and `hold` (reuse the previous latent). For every prompt and segment we generate both modes from the same boundary and the same noise, so the only difference is the generator; a segment is *coarse-unsafe* when the coarse segment is farther from the four-step fine reference than the fine segment. Costs are wall-clock DiT seconds with device synchronisation.

IV. Protocol, Artifacts and Reproducibility

Every experiment is reproducible from stored artifacts without a GPU. One audit script under `jevgag/code/` produces one result bundle under `jevgag/results/`, and the claim-to-artifact audit (§V-I) re-derives each headline number from those files.

- **E1 (cost).** `e1_breakeven.py` → `results/e1_breakeven/`: a ten-cell matched matrix over {T4, A100} × {fine, coarse} schedules.
- **E2 (signal).** `e2_image_signal.py`, `e2_text14.py`, `e2_hybrid.py`, `e2_audit.py` → `results/e2_signal/`: 80 paired segments.
- **E3 (operating point).** `e3_online_matched.py` → `results/e3_online/`: 12 prompts on A100.
- **E4 (controller).** `e4_rq3_seam.py` → `results/e4_controller/`: three controllers on two GPUs.
- **E5 (endpoint).** `e5_endpoint.py`, `e5_audit.py` → `results/e5_endpoint/`: 48 videos over six policies.
- **E6 (T4 boundary).** `e6_t4_boundary.py` → `results/e6_t4_boundary/`: three fine/coarse pairs.
- **E7 (confidence).** `e7_confidence_gate.py` → `results/e7_confidence/`: three budgets.
- **E8 (recalibration).** `e8_audit.py`, `e8_figure.py` → `results/e8_recalibration/`: n=12 (A100), n=8 (T4).
- **E9 (audit).** `e9_claim_audit.py` → `results/e9_audit/`: 18 claims.

Figures are regenerated by `code/e_figures.py`, and the off-device kernels that produced the A100 results are pinned under `jevgag/compute/` (`colab/`, `jobs/`). The result bundles are mirrored as public Kaggle datasets.

V. Results

A. E1 — Break-even is hardware- and operating-point-conditional

Table I reports the matched matrix; Fig. 1 visualises it. On T4 the cheap coarse `c3-3 s1` beats fine only when fine uses ≥ 2 steps, and the expensive `c3-3 s4` is *never* cheaper; on A100 the cheap coarse beats all three fine modes. The same pair (`fine s2`, `coarse s2`) therefore lies on opposite sides of Eq. (1) on the two GPUs (2.97 vs. break-even 2.68 on T4; 1.74 on A100). This is the confound resolved: the prior audit compared expensive coarse on T4 with cheap coarse on A100.

TABLE I
E1 — matched break-even matrix. *Ratio* is the measured per-forward ratio; coarse is cheaper iff ratio $< \rho_{\mathrm{BE}}$ (Eq. (1)).

GPU	fine / coarse	NFE f/c	ρ_{be}	ratio	cheaper
T4	s_1 / s_1	19.0/8.33	2.28	2.694	no
T4	s_1 / s_4	19.0/21.33	0.891	2.197	no
T4	s_2 / s_1	34.0/8.33	4.08	3.301	**yes**
T4	s_2 / s_2	34.0/12.67	2.684	2.969	no
T4	s_4 / s_1	64.0/8.33	7.68	3.841	**yes**
T4	s_4 / s_4	64.0/21.33	3.0	3.133	no
A100	s_1 / s_1	19.0/8.33	2.28	1.561	**yes**
A100	s_2 / s_1	34.0/8.33	4.08	1.95	**yes**
A100	s_2 / s_2	34.0/12.67	2.684	1.744	**yes**
A100	s_4 / s_4	64.0/21.33	3.0	1.84	**yes**

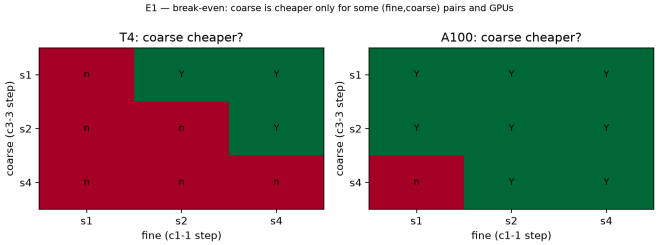


Fig. 1. E1 — break-even over hardware and operating point. Coarse is cheaper only where the measured ratio falls below the schedule break-even, and the same pair can sit on opposite sides on the two GPUs.

B. E2 — Image-grounded Jev does not beat text Jev as a signal

On 80 matched segments (Table II) the image and hybrid states are *worse* than the text state: paired ΔAUROC of −0.110 (difficulty), −0.108 (drift) and −0.195 (motion), the last with a 95 % CI excluding zero ($P(\text{image} > \text{text}) = 0.006$). Text-1.13 ≈ text-1.14, so the deficit is the *modality*, not the model version. This is an honest negative and it is consistent with

independent evidence that pixel-only quality prediction with such models is unreliable [17].

TABLE II
E2 — SIGNAL AUDIT ON 80 PAIRED SEGMENTS (AUROC). IMAGE AND HYBRID DO NOT BEAT TEXT.

signal	text-1.13	text-1.14	image-1.14	hybrid-1.14
Jev difficulty	0.788	0.777	0.667	0.687
Jev drift	0.738	0.755	0.647	0.656
Jev motion	0.703	0.706	0.510	0.568
warp residual	0.586	0.586	0.586	0.586
motion stat	0.607	0.607	0.607	0.607
complexity	0.368	0.368	0.368	0.368

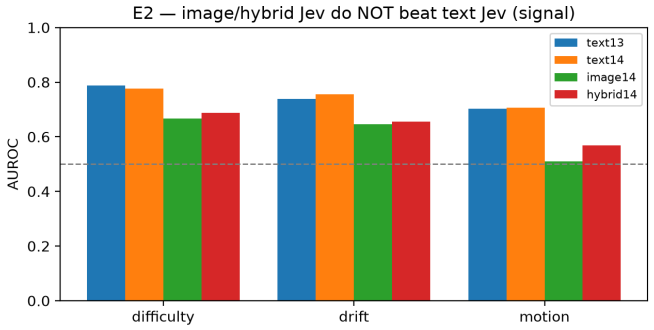


Fig. 2. E2 — image- and hybrid-grounded Jev do not beat text-grounded Jev as a routing signal; the motion deficit is significant ($P = 0.006$).

C. E3 — The adaptive outcome flips with the operating point alone

Same A100, same 12 prompts, only the coarse step differs (Table III). With the *expensive* coarse the online policy is forced to 92 % fine and cannot beat fixed fine (4.43 s vs. 3.81 s); with the *cheap* coarse it uses 21 % fine and is both cheaper and closer to the reference (2.55 s vs. 3.95 s). The adaptive verdict is therefore a property of the operating point, not of the signal alone.

TABLE III
E3 — MATCHED A100 ONLINE FACTORIAL (SAME GPU, SAME 12 PROMPTS). ONLY THE COARSE STEP CHANGES.

	expensive coarse s_4			cheap coarse s_1		
policy	DiT (s)	refL1	fine%	DiT (s)	refL1	fine%
fixed fine	3.81	0.2812	1.00	3.95	0.2832	1.00
fixed coarse	4.43	0.2874	0.00	2.55	0.2728	0.00
fixed hold	0.76	0.2824	0.00	0.77	0.2674	0.00
random 50%	4.12	0.2868	0.50	3.24	0.2789	0.50
adaptive warp	3.87	0.2820	0.92	2.85	0.2733	0.21

D. E4 — Recovery, hysteresis and the switch seam

Table IV reports the controller ablation. Adding a hysteresis band and a forced-recovery window reduces switches on both GPUs without worsening the area-under-horizon regret, and a switch *seam* is measurable: the switch-adjacent L_1 exceeds the interior L_1 (ratio 1.18 on T4, 1.98 on A100). The seam is the cost of switching, and it is a target for the companion method.

TABLE IV
E4 — CONTROLLER ABLATION. SWITCHES FALL WITHOUT WORSENING AUH REGRET; A SEAM EXISTS.

GPU	controller	switches	fine frac	AUH refL1	seam ratio
T4	threshold only	0.50	0.50	0.3133	1.30
T4	full (hyst.+rec.)	0.38	0.53	0.3116	1.18
T4	full, noisy	0.62	0.66	0.3150	1.53
A100	threshold only	0.50	0.50	0.2767	2.20
A100	full (hyst.+rec.)	0.42	0.52	0.2772	1.98
A100	full, noisy	0.50	0.56	0.2718	1.75

E. E5 — Image-grounded Jev is a validated independent endpoint

Across six policies and 48 videos, the image-grounded endpoint agrees with CLIP subject consistency [14] at Spearman ρ=+1.000 (same-subject) and +0.943 (consistency); Table V lists the per-policy values and Fig. 3 plots them. This is the report's strongest positive: the image oracle cannot *drive* the decision (E2) but it can *certify* it, which is exactly what P3 requires.

TABLE V
E5 — IMAGE-GROUNDED JEV ENDPOINT VS. CLIP SUBJECT CONSISTENCY (48 VIDEOS, 6 POLICIES).

policy	Jev consistency	Jev same-subject	CLIP subj. cons.
reference	3.461	0.894	0.990
fixed fine	2.151	0.416	0.984
adaptive warp	2.081	0.410	0.980
random 50%	1.978	0.286	0.965
alternating	1.626	0.268	0.959
fixed coarse	1.775	0.230	0.949

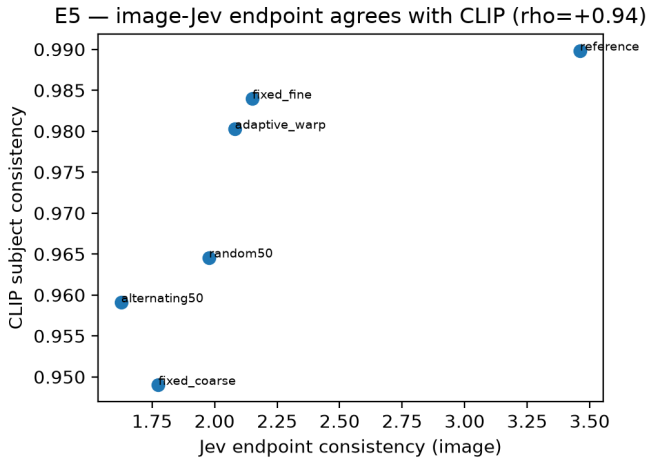


Fig. 3. E5 — the image-grounded Jev endpoint agrees with CLIP subject consistency at $\rho = +1.000$ (same-subject) and +0.943 (consistency).

F. E6 — The T4 negative is an operating-point boundary, not a signal failure

With fine `c1-1 s1`, the cheap coarse `c3-3 s1` saves −4.57 s on T4, i.e. it is *slower*, so adaptive routing cannot save wall-clock time regardless of how good the signal is. Coarse only wins on T4 once fine uses $\geq$ 2 steps (+7.01 s at s_2,

+29.70 s at s_4). This mirrors E1 and completes the confound explanation.

G. E7 — Confidence-gating is neutral

Gating on the oracle's own confidence does not improve the regret–cost frontier (Table VI): the confidence-gated policy differs from the difficulty-only policy by $|\Delta| \leq 0.0013$ at every budget (25 %, 50 %, 75 % fine). Honest negative.

TABLE VI
E7 — CONFIDENCE-GATED ROUTING FRONTIER (MEAN REFERENCE REGRET; LOWER IS BETTER).

policy	25 % fine	50 % fine	75 % fine
random	0.3044	0.2957	0.2881
difficulty only	0.2968	0.2875	0.2781
confidence gated	0.2964	0.2875	0.2794
oracle	0.2734	0.2734	0.2734

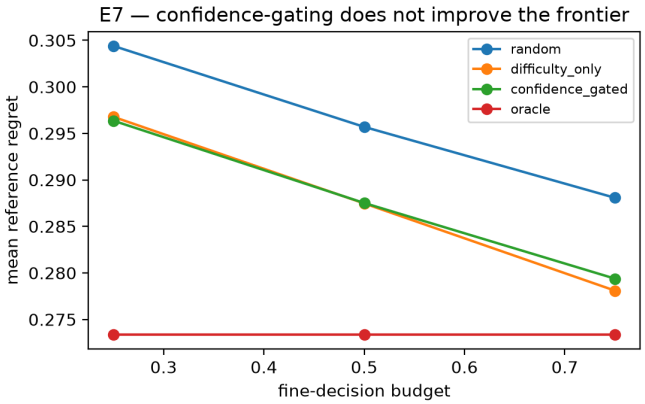


Fig. 4. E7 — confidence gating is neutral: the frontier is indistinguishable from the difficulty-only policy at every budget.

H. E8 — Stability-gated recalibration does not help

Injecting a high-step fine segment when drift is detected is worse on both GPUs (Table VII, Fig. 5): on A100 recalibration minus full gives refL1 +0.0112 at DiT +2.04 s; on T4 recalibration minus adaptive_warp gives refL1 +0.0202 at DiT +12.73 s. Honest negative.

TABLE VII
E8 — STABILITY-GATED RECALIBRATION. LEFT: A100 ($n = 12$). RIGHT: T4 ($n = 8$).

A100 ($n = 12$)		T4 ($n = 8$)	
controller	refL1	policy	refL1
threshold only	0.2767	fixed coarse	0.3186
full	0.2772	fixed fine	0.2998
recalibrate	0.2884	adaptive warp	0.2949
		recalibrate	0.3151

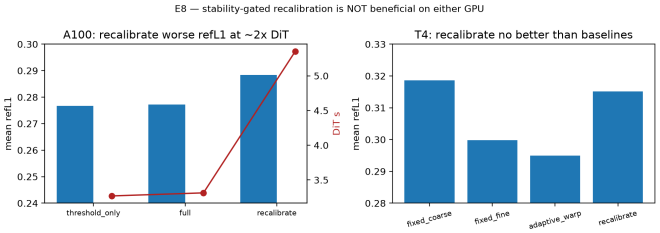


Fig. 5. E8 — recalibration is not beneficial on either GPU: it raises the endpoint distance at higher cost.

I. E9 — Claim-to-artifact audit

Every headline number in this report is re-derived from a stored artifact. The audit (`jevgag/results/e9_audit/claim_audit.json`) reports **18/18 PASS**, covering E1 (3 claims), E2 (5), E3 (3), E4 (4), E5 (2) and E7 (1). Table VIII summarises the coverage.

TABLE VIII
E9 — CLAIM-TO-ARTIFACT AUDIT COVERAGE (18/18 PASS).

experiment	claims audited	example	result
E1	3	T4 `c3-3 s4` never cheaper	PASS
E2	5	motion deficit significant ($P = 0.006$)	PASS
E3	3	cheap coarse cheaper online on A100	PASS
E4	4	switches 0.50 → 0.38 (T4)	PASS
E5	2	endpoint $\rho = +1.000$	PASS
E7	1	confidence gating neutral	PASS
		total	**18/18**

VI. Discussion and Limitations

a) What the report establishes.: The image-grounded oracle is a *poor routing signal* but an *excellent independent endpoint*; the break-even law (Eq. (1)) is the reusable engineering tool; and the adaptive verdict is conditional on hardware *and* operating point, which explains the earlier confound.

b) Limitations.: (i) *One backbone*: all experiments use a single causal distilled model, so the numbers do not transfer to other generators. (ii) *Two accelerators and one schedule family*: T4 and A100-80GB only. (iii) *Small samples*: the signal test uses 80 paired segments and the recalibration arms use 8–12 prompts, so the intervals are wide. (iv) *Off-device dependency*: the oracle is a versioned external service (we pin Jev 1.14) and the results depend on its behaviour; the companion method removes this dependency. (v) *One modality comparison*: text vs. image; other modalities (audio, depth) are untested. (vi) *No step-allocation or chunk-shape baseline*: the closest published alternatives are discussed but not re-implemented.

c) Threats to validity.: The endpoint result (E5) rests on 48 videos from six policies, all on T4; a larger, multi-GPU endpoint study would strengthen it. The recalibration negatives (E8) are measured on a single seed.

VII. Conclusion

An off-device, image-grounded System One model can *certify* adaptive temporal granularity even when it cannot *drive* it, and the cost side obeys a matched break-even law over

hardware and operating point. The routing signal remains the open problem, and we report the two failed policies openly rather than tuning them to look better.

DECLARATIONS

Competing interests: none. **Funding:** none. **Ethics:** not applicable (no human or animal subjects). **Data availability:** all code, figures and audited result files are public at https://github.com/j1s4nn/JevGAG; the result bundles are published as public Kaggle datasets. **CRediT:** Hossen Md Jisan — Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing — original draft, Writing — review and editing.

REFERENCES

[1] M. J. Hossen, "Jev-routed temporal granularity with recovery and hysteresis for causal video generation," Preprint, 2026. [Online]. Available: https://doi.org/10.5281/zenodo.23395414

[2] X. Huang, Z. Li, G. He, M. Zhou, and E. Shechtman, "Self forcing: Bridging the train-test gap in autoregressive video diffusion," 2025. [Online]. Available: https://arxiv.org/abs/2506.08009

[3] H. Zhu, M. Zhao, G. He, H. Su, C. Li, and J. Zhu, "Causal forcing: Autoregressive diffusion distillation done right for high-quality real-time interactive video generation," 2026. [Online]. Available: https://arxiv.org/abs/2602.02214

[4] K. Zheng, G. He, M. Zhao, J. Zhang, H. Chen, J. Chen, C.-H. Lin, M.-Y. Liu, J. Zhu, and Q. Ma, "Causal-rcm: A unified teacher-forcing and self-forcing open recipe for autoregressive diffusion distillation in streaming video generation and interactive world models," 2026. [Online]. Available: https://arxiv.org/abs/2606.25473

[5] A. Graves, "Adaptive computation time for recurrent neural networks," 2016. [Online]. Available: https://arxiv.org/abs/1603.08983

[6] S. Teerapittayanon, B. McDanel, and H. T. Kung, "Branchynet: Fast inference via early exiting from deep neural networks," 2017. [Online]. Available: https://arxiv.org/abs/1709.01686

[7] T. Sun, Y. Zhou, X. Liu, X. Zhang, H. Jiang, Z. Cao, X. Huang, and X. Qiu, "Early exiting with ensemble internal classifiers," 2021. [Online]. Available: https://arxiv.org/abs/2105.13792

[8] T.-T. Le, Y. Zhao, M. Chai, Z. Shen, Z. Cao, D. Tang, X. Xie, and D. Kong, "Dsa: Dynamic step allocation for fast autoregressive video generation," 2026. [Online]. Available: https://arxiv.org/abs/2606.04432

[9] X. Ma, J. Berner, C. Liu, A. Vahdat, W. Nie, and X. Wang, "Flex-forcing: Towards a unified autoregressive and bidirectional video diffusion model," 2026. [Online]. Available: https://arxiv.org/abs/2607.03509

[10] Y. Wang, Y. Wang, T. Xue, Y. Qiao, Y. Wang, X. Chen, and C. Xu, "Pare: Pruning and adaptive routing for efficient video generation," 2026. [Online]. Available: https://arxiv.org/abs/2605.27336

[11] M. Adnan, N. Kurella, A. Arunkumar, and P. J. Nair, "Foresight: Adaptive layer reuse for accelerated and high-quality text-to-video generation," 2025. [Online]. Available: https://arxiv.org/abs/2506.00329

[12] Z. Huang, Y. He, J. Yu, F. Zhang, C. Si, Y. Jiang, Y. Zhang, T. Wu, Q. Jin, N. Chanpaisit, Y. Wang, X. Chen, L. Wang, D. Lin, Y. Qiao, and Z. Liu, "Vbench: Comprehensive benchmark suite for video generative models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 21 807–21 818.

[13] D. Zheng, Z. Huang, H. Liu, K. Zou, Y. He, F. Zhang, L. Gu, Y. Zhang, J. He, W.-S. Zheng, Y. Qiao, and Z. Liu, "Vbench-2.0: Advancing video generation benchmark suite for intrinsic faithfulness," 2025. [Online]. Available: https://arxiv.org/abs/2503.21755

[14] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021, pp. 8748–8763.

[15] J. Li, D. Li, C. Xiong, and S. Hoi, "Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022, pp. 12 888–12 900.

[16] G. Ling, M. Xue, and Z. Ye, "Jev in the wild: A data-driven analysis of the jev model's functionality, applications and ecosystem," 2026. [Online]. Available: https://arxiv.org/abs/2609.30216

[17] W. Robitza, "Jevqa - video quality from metadata, bitstream, and pixel features with a general-purpose decision model," 2026. [Online]. Available: https://arxiv.org/abs/2609.24395